ASSIGNMENT TWO: DUE DATE 21^{st} NOVEMBER 2024

1) A factory produces defective items in a batch, where the number of defective items Y in a randomly selected batch follows a discrete probability distribution. The probability mass function for the number of defective items is given by:

$$f(y) = \frac{6-y}{15}, y = 1, 2, 3, 4, 5.$$

Where y represents the number of defective items in a batch

i. Verify that $f(y)$ is a valid probability mass function **[2 marks]**

ii. Find the probability that a randomly selected batch contains at least 3 defective items. **[3 marks]**

iii. Suppose the factory monitors batches and rejects any batch with more than 4 defective items. What is the probability of rejecting a randomly selected batch. **[2marks]**

iv. Calculate the expected number of defective items, E(Y). **[3 marks]**

v. Calculate the variance of the number of defective items. Var(Y) **[3 marks]**

2) A company analyzed online shopping behavior and found that:
- 75% of customers browse the website (W)
- 65% make a purchase (P)
- 50% of customers do both, browse the website and make a purchase.

If a customer is selected at random, what is the probability that:

i. The customer either browses the website or makes a purchase (or both)? **[2 marks]**

ii. The customer browses the website but does not make a purchase? **[2 marks]**

iii. The customer makes a purchase but does not browse the website? **[2 marks]**

3) A factory produces a batch of 30 electronic components, 12 of which are defective and 18 are non-defective. If a quality control inspector randomly selects 8 components for testing, what is the probability that:

i. At least 5 of the selected components are non-defective? **[3 marks]**

ii. At most 2 of the selected components are defective? **[3 marks]**

4) An airport uses three baggage handling systems (A, B, and C), which process 50%, 30%, and 20% of the total baggage, respectively. The probabilities of losing a piece of baggage are:

- System A: 0.1%,
- System B: 0.2%,
- System C: 0.3%.

i. Calculate the probability that a randomly selected baggage is lost. **[2 marks]**

ii. If a lost baggage is found, what is the probability that it was processed by System C? **[3 marks]**